

Heisenberg Groups

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1 The CR Structures on the Heisenberg Group

Definition 1.1 (cf. [?]). The **Heisenberg group** (of degree n) is the Lie group H_n whose underlying manifold is $\mathbb{C}^n \times \mathbb{R}$ with coordinates $(z_1, \dots, z_n, t) \equiv (z, t)$ and whose group law is given by

$$(z, t) \cdot (z', t') = (z + z', t + t' + 2\Im(z \cdot z')), \quad z \cdot z' = \sum_{i=1}^n z_i \bar{z}'_i.$$

If we set $z_j = x_j + iy_j$, then $(x_1, \dots, x_n, y_1, \dots, y_n, t)$ form a real coordinate system for H_n . In this coordinate system we define the following vector fields:

$$X_j = \frac{\partial}{\partial x_j} + 2y_j \frac{\partial}{\partial t}, \quad Y_j = \frac{\partial}{\partial y_j} - 2x_j \frac{\partial}{\partial t}, \quad T = \frac{\partial}{\partial t} \quad (1)$$

It is a simple exercise to verify that $\{X_1, \dots, X_n, Y_1, \dots, Y_n, T\}$ is a basis for the **left-invariant vector fields** on H_n . The commutation relations are as follows:

$$[Y_j, X_k] = 4\delta_{jk}T, \quad [X_j, X_k] = [Y_j, Y_k] = [X_j, T] = [Y_j, T] = 0. \quad (2)$$

Remark 1.1. Note that these are an abstract form of the commutation relations for the quantum-mechanical position and momentum operators in n -space; hence the name “Heisenberg group”.

Remark 1.2. If one use

$$X_j = \frac{\partial}{\partial x_j} + \frac{1}{2}y_j \frac{\partial}{\partial t}$$

to define the distribution, then they are not left-invariant. And the Levi form τ defined below has to be changed.

We further define the complex vector fields

$$Z_j = \frac{1}{2}(X_j - iY_j) = \frac{\partial}{\partial z_j} + i\bar{z}_j \frac{\partial}{\partial t}, \quad \bar{Z}_j = \frac{1}{2}(X_j + iY_j) = \frac{\partial}{\partial \bar{z}_j} - iz_j \frac{\partial}{\partial t} \quad (3)$$

whose commutation relations are

$$[Z_j, \bar{Z}_k] = -2i\delta_{jk}T, \quad [Z_j, Z_k] = [\bar{Z}_j, \bar{Z}_k] = [Z_j, T] = [\bar{Z}_j, T] = 0, \quad (4)$$

and the one-form

$$\tau = dt + 2 \sum_{j=1}^n (x_j dy_j - y_j dx_j). \quad (5)$$

We note that $\langle \tau, Z_j \rangle = \langle \tau, \bar{Z}_j \rangle = 0$ for all j . It is immediate that the subbundle $T^{1,0}$ of $\mathbb{C}TH_n$ spanned by $Z_1, \dots, Z_n$ defines a CR structure on H_n . To compute the Lévi form, we have

$$\langle Z_j, Z_k \rangle_L = \frac{1}{2i} \langle \tau, [Z_j, \bar{Z}_k] \rangle = \frac{1}{2i} \langle \tau, -2i\delta_{jk}T \rangle = \delta_{jk}.$$

Thus H_n is **strongly pseudoconvex**.

Remark 1.3. (??) is equal to (cf. [?])

$$\theta_0 = dt + i \sum_{j=1}^n (z^j d\bar{z}^j - \bar{z}^j dz^j) \quad (6)$$

in complex form, and by differentiating it we obtain

$$d\theta_0 = 2i \sum_{j=1}^n dz^j \wedge d\bar{z}^j. \quad (7)$$

1.1 A geometrical interpretation

This has the following geometrical interpretation due to Pyatetski-Shapiro. Set

$$D_{n+1} = \left\{ \zeta \in \mathbb{C}^{n+1} : \sum_{j=1}^n |\zeta_j|^2 - \Im \zeta_0 < 0 \right\},$$

$$M_n = \partial D_{n+1} = \left\{ \zeta \in \mathbb{C}^{n+1} : \sum_{j=1}^n |\zeta_j|^2 = \Im \zeta_0 \right\}.$$

D_{n+1} (known as the Siegel domain) is holomorphically equivalent to the unit ball in $\mathbb{C}^{n+1}$. H_n acts on $\mathbb{C}^{n+1}$ by holomorphic affine transformations which preserve D_{n+1} and M_n , as follows: if $(z, t) \in H_n$ and $\zeta \in \mathbb{C}^{n+1}$, $(z, t) \cdot \zeta = \zeta'$, where

$$\zeta'_0 = \zeta_0 + t + i|z|^2 + 2i \sum_{j=1}^n \zeta_j \bar{z}_j,$$

$$\zeta'_j = \zeta_j + z_j, \quad j > 0.$$

H_n acts simply transitively on M_n and so may be identified with M_n via the correspondence

$$(z, t) \longleftrightarrow (z, t) \cdot 0 = (t + i|z|^2, z_1, \dots, z_n).$$

Under this identification the CR structure on H_n described above coincides with the CR structure on M_n induced from $\mathbb{C}^{n+1}$. (Indeed, one easily checks that the holomorphic subspaces at the origin are the same since M_n is tangent to the hyperplane $\zeta_0 = 0$ there. But the CR structure on H_n is left-invariant, and the action of H_n on M_n , being holomorphic, preserves the CR structure on M_n ; hence the holomorphic subspaces coincide everywhere.)

Remark 1.4. The action is **simply transitive** (or sharply transitive, or regular) if it is both transitive and free (cf. [wikipedia](#)).

1.2 Another useful identification

Another useful identification is that of the Heisenberg group and the sphere $S^{2n+1} \subset \mathbb{C}^{n+1}$ minus a point.

Let $\Omega \subset \mathbb{C}^N$ be a domain with smooth boundary $\partial\Omega$, i.e., there is an open neighborhood U of the closure $\bar{\Omega}$ in $\mathbb{C}$ and a C^∞ function $\rho: U \rightarrow \mathbb{R}$ such that $\Omega = \{x \in U: \rho(x) > 0\}$ (and $\partial\Omega = \{x \in U: \rho(x) = 0\}$) and $(D\rho)_x \neq 0$ at any $x \in \partial\Omega$. Let $T_{1,0}(\partial\Omega)$ be the induced CR structure on $\partial\Omega$, as a real hypersurface in $\mathbb{C}^N$. Let θ be the pullback to $\partial\Omega$ of the real 1-form $i(\bar{\partial} - \partial)\rho$ on U . Then θ is a pseudo-Hermitian structure on $(\partial\Omega, T_{1,0}(\partial\Omega))$. As we just saw, the boundary of the Siegel domain is a strictly pseudoconvex CR manifold.

Proof. From

$$\partial\rho = \sum_{j=1}^{n+1} \frac{1}{2} \left(\frac{\partial}{\partial x_j} - i \frac{\partial}{\partial y_j} \right) \rho (dx_j + i dy_j)$$

we know

$$i(\bar{\partial} - \partial)\rho = \frac{\partial\rho}{\partial x_j} dy_j - \frac{\partial\rho}{\partial y_j} dx_j.$$

Now it's easy to verify

$$\theta(J(\nabla\rho)) = |\nabla\rho|^2 \neq 0, \quad \theta(T_j) = 0, \forall T_j \perp \{\nabla\rho, J\nabla\rho\}.$$

□

Also, the sphere $S^{2n+1} \subset \mathbb{C}^{n+1}$ is a strictly pseudoconvex manifold, since the boundary of the unit ball $B_{n+1} = \{z \in \mathbb{C}^{n+1}: |z| < 1\}$, and the (restriction to $S^{2n+1} \setminus \{p\}$ of the) Cayley transform (cf. Friedrich Haslinger's notes.)

$$\Phi(z_1, \dots, z_n, z_{n+1}) = (w_1, \dots, w_n, w_{n+1}),$$

where $w_j = \frac{z_j}{(1+z_n)}$ for $1 \leq j \leq n$ and $w_{n+1} = i \frac{(1-z_{n+1})}{(1+z_{n+1})}$. This gives a CR isomorphism $S^{2n+1} \setminus \{p\} \cong \partial\Omega_{2n+1}$ (and thus a CR isomorphism $S^{2n+1} \setminus \{p\} \cong H_n$).

1.3 A left-invariant k -strongly pseudoconvex CR structure

We can also define a left-invariant CR structure on H_n which is k -strongly pseudoconvex, $\frac{n}{2} \leq k \leq n$, by taking $T_{1,0}$ to be the bundle spanned by $Z_1, \dots, Z_k, \bar{Z}_{k+1}, \dots, \bar{Z}_n$. A few words of explanation about this will be helpful. Let us take $(w_1, \dots, w_n, t)$ as new coordinates on H_n , where $w_j = z_j, 1 \leq j \leq k$, and $w_j = \bar{z}_j, k+1 \leq j \leq n$. Set $\varepsilon_j = 1$ or -1 according as $j \leq k$ or $j > k$, and define the Hermitian form Q_k on $\mathbb{C}^n$ by

$$Q_k(w, w') = \sum_{j=1}^n \varepsilon_j w_j \bar{w}'_j.$$

Then in the coordinates (w, t) the group law is

$$(w, t)(w', t') = (w + w', t + t' + 2\Im Q_k(w, w')).$$

The left-invariant vector field agreeing with $\frac{\partial}{\partial w_j}$ at the origin is

$$W_j = \frac{\partial}{\partial w_j} + \varepsilon_j i \bar{w}_j \frac{\partial}{\partial t}, \quad (8)$$

and $T_{1,0}$ is the bundle spanned by $W_1, \dots, W_n$. In fact, $W_j = Z_j, j \leq k$, and $W_j = \bar{Z}_j, j > k$. As above, the Levi form is given by $\langle W_j, W_l \rangle_L = \varepsilon_j \delta_{jl}$. The geometrical realization of H_n also has an analogue here: H_n may be embedded in $\mathbb{C}^{n+1}$ as the hypersurface

$$\{\zeta \in \mathbb{C}^{n+1} : \Im \zeta_0 = Q_k(\zeta', \zeta')\}, \quad \zeta' = (\zeta_1, \dots, \zeta_n), \quad (9)$$

via the mapping

$$(w, t) \mapsto (t + iQ_k(w, w), w_1, \dots, w_n)$$

which preserves the CR structure.

1.4 Unsolvability Operators

At this point we should also mention that, in the case $n = 1$, the vector field $\bar{Z} = (\partial/\partial z) - iz(\partial/\partial t)$ which spans $T_{0,1}$ is the unsolvable operator of H. Lewy. Indeed, Lewy was led to this phenomenon by considering boundary values of holomorphic functions on D_2 .

Theorem 1.2 (cf. [?]). *Let L denote the differential operator acting on functions $u(x, y, z)$ defined by*

$$Lu = -u_x - iu_y + 2i(x + iy)u_z. \quad (1.1a)$$

There exists a function $F(x, y, z) \in C^\infty(\mathbb{R}^3)$ such that the equation

$$Lu = F(x, y, z) \quad (1.1b)$$

has no solution whose domain is an open set Ω in $\mathbb{R}^3$, with $u \in C^1(\Omega)$, and u_x, u_y, u_z Hölder continuous in Ω .